VITAL: An End-to-End Contactless Vitals-to-Triage Pipeline with Hybrid rPPG Correction and a Measured Case for Deterministic Clinical Safety

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***Abstract*—Contactless heart-rate sensing from a webcam is usually evaluated on its own, yet it is increasingly paired with language models trusted to reason about clinical risk straight from raw vitals. VITAL tests both assumptions together. Its rPPG pipeline – CHROM corrected by a small gradient-boosted model – reaches 6.48 BPM mean absolute error across 71 real sessions from UBFC-rPPG and UBFC-Phys, down from 9.43 BPM classically – an aggregate gain concentrated in a subset of hard sessions, not a uniform per-subject effect, landing mostly where three DSP fixes could not reach: motion artifact from talking. Separately, we asked whether VITAL's language-model narration layer can be trusted to simply repeat a number it was handed. Across 16 runs on 8 controlled clinical scenarios, it echoed the correct priority score verbatim only 6.2% of the time, even when explicitly told not to recompute it. That result is why a deterministic scoring engine, not the language model, remains the sole source of truth for acuity, shock status, and priority in VITAL. We report what worked, what failed, and where this system still trails the closest published rPPG benchmark.**

***Index Terms*—Remote photoplethysmography, rPPG, CHROM, hybrid machine learning, emergency triage, clinical decision support, multi-agent LLM systems, deterministic safety, contactless vitals monitoring**

I. Introduction

A. Background and Motivation

Emergency departments triage patients under time pressure with a static, sensor-dependent workflow, and every minute spent attaching a sensor to a waiting patient is a minute deterioration can go unnoticed. **Remote photoplethysmography (rPPG)** sidesteps this: heart rate, HRV, and respiratory rate can be pulled from an ordinary webcam feed with no contact at all [2], [3], [9]. Meanwhile, **large language models** are increasingly proposed as the reasoning layer that turns those vitals into a triage decision, often through multi-agent frameworks splitting perception, diagnosis, and coordination across separate LLM agents [14], [15]. It is an attractive pairing – a webcam and an LLM API in place of a nurse's stopwatch and a static rule table – but it raises a question rarely measured directly: when an LLM agent is handed a patient's exact clinical numbers and told only to restate them, **does it actually do so?**

B. Problem Statement

Two gaps motivate this work. The first is methodological: rPPG systems are often validated once, against a clean benchmark signal, without the harder work of tracing why specific real subjects fail and fixing the cause rather than tuning a global threshold. The second is more consequential – agentic-AI clinical tools increasingly let an LLM see raw vitals and reason toward a severity classification directly [13], without checking whether it reliably preserves a number it was only meant to **restate, not recompute**. Warrier and Abhilash's agentic orchestration framework for hospital workflows [14] builds in a runtime assurance layer to veto unsafe agent actions – a guardrail their architecture addresses procedurally, **not empirically**.

C. Contributions

This paper makes five contributions, each backed by a measured result:

1) A complete, deployed contactless-vitals-to-triage pipeline – webcam capture, **MediaPipe** ROI extraction, a **CHROM** rPPG engine, a **deterministic clinical scoring engine**, a three-agent **CrewAI** narration layer, and a Next.js/Flask/Neon Postgres stack – running end to end, not a benchmarking script.

2) **Three independent classical-DSP bugs** in the rPPG engine, each root-caused by tracing a specific failing subject rather than tuning a global parameter: a harmonic lock-on bug, a hardcoded-frame-rate bug, and a history-aggregation freeze bug. Combined, they took the real-video baseline from **17.4 to 8.4 BPM MAE** on the original 56-session set.
3) A **hybrid classical-DSP + gradient-boosted correction layer**, validated with subject-level **GroupKFold**, reducing real-video MAE from **9.43 to 6.48 BPM (31% relative reduction)** across **71 sessions and 51 subjects** – concentrated in a subset of hard cases (Section VI), largest in speech, the one condition DSP-level fixes never touched.
4) A deterministic scoring engine expressing **Compensated Shock** as a **multiplicative interaction** between tachycardia and tachypnea ramps rather than a boolean AND-gate, with an explicitly enforced architectural rule that this engine alone is the **sole source of truth** for the numbers driving triage priority.
5) A direct, repeated measurement of whether an LLM agent faithfully restates a given clinical number: across **16 real CrewAI runs**, the priority score was restated correctly only **6.2% of the time**, even under an explicit "do not recompute" instruction – turning the guardrail need identified in [14] into a concrete number.

II. Related Work

A. rPPG Signal Extraction

CHROM [2] and **POS** [3] remain the two dominant classical rPPG projection methods. Pirzada et al.'s **2024 survey of over 300 rPPG studies** found both still sensitive to motion artifact and lighting change, with the field mostly validated in labs, not realistic settings [9] – the same sensitivity traced here in the speech-condition failures. A **2026 clinical validation** of a commercial system in 47 cardiovascular patients reports **1.06 BPM MAE** against ECG [10]; that gap is a large part of why this paper does not overstate its own.

B. ML-Corrected and Hybrid rPPG

The closest published comparator to VITAL's hybrid layer is **Qaadan et al.**, combining five rPPG methods with four classical regressors across four datasets under a single 80/20 split [1]. Their best result – CHROM into XGBoost with permutation-importance feature reduction – reaches **4.60 BPM MAE / 5.57 RMSE**, ahead of this project's comparable numbers (Section VI). Their conclusion names **"hybrid algorithmic-learned models"** as future work they never built [1]; that is exactly VITAL's architecture.

C. Agentic and Learning-Based Emergency Triage

A 2025 review of interpretable ML for ED triage found **gradient-boosted models** dominate (AUROC 0.78–0.96), flagging gaps in operational-metric evaluation and fairness [11]. Yaddaden et al.'s **Polynomial Features transform** reaches **99.80% accuracy** for ESI prediction on MIMIC-IV [12] – the same idea applied here, multiplying a tachycardia ramp by a tachypnea ramp. **Triage-Bot** pairs an rPPG extractor with a rule-based CTAS classifier [15], evaluation still pending. Liu and Tsai pair explainability with a **RAG pipeline** [13]; VITAL's retrieval is simpler, term-frequency cosine similarity. None checks whether an LLM layer's output **reproduces a clinical number faithfully**.

D. Research Gap

No system we reviewed combines root-caused DSP fixes, a subject-level cross-validated correction layer, an enforced deterministic/LLM boundary, and a repeated faithfulness measurement. The correction layer closes a gap Qaadan et al. name as unbuilt future work [1]; the faithfulness measurement turns a guardrail need the agentic-orchestration literature addresses only procedurally [14] into an actual number.

III. System Architecture

A. Overview

VITAL runs as **three layers** on a live Flask API and Next.js dashboard, backed by Neon Postgres. A **perception layer** extracts a face ROI via MediaPipe Face Mesh (Haar-cascade fallback) and feeds normalized RGB means into an rPPG DSP engine. A **clinical-reasoning layer** deterministically computes a custom ESI-style acuity score (Section III-D), a Compensated Shock flag, and a priority score. A **narration layer** – three CrewAI agents – restates that output for the dashboard and ARIA, a voice assistant with read-only database access. Fig. III summarizes the pipeline.

B. Perception Layer

MediaPipe's [6] **468-point Face Mesh** isolates forehead and malar-cheek landmarks with yaw-aware gating, so an off-angle cheek region is excluded rather than blended in noisily. The layer also estimates **ambient luminance** and **centroid drift** per frame, both exposed to the correction model as candidate features.

C. rPPG DSP Engine

The default projection is **CHROM** [2]: per-channel normalization, projection onto two chrominance axes, and adaptive scaling by the projected-signal standard-deviation ratio. A **Welch periodogram** locates the dominant heart-rate frequency; **cubic-spline upsampling to 250 Hz** supports RMSSD-based HRV; a 0.10-0.50 Hz bandpass extracts respiratory rate. **POS** [3] is a hot-swappable alternative, **not wired into any live route** (Section V-C).

D. Deterministic Clinical Engine

The **deterministic clinical scoring module** computes **ESI-style acuity**, the **Compensated Shock flag**, and a **1-100 priority score** directly from extracted vitals in plain code – never sampled from an LLM. Rather than a boolean gate that simply checks whether heart rate exceeds 100 bpm and

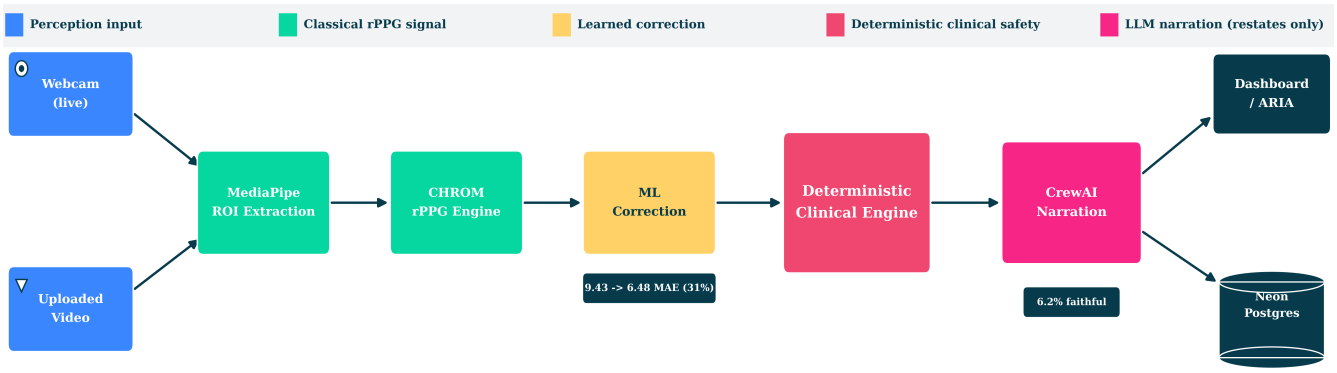


Fig. 1. VITAL's end-to-end pipeline. Blue: perception input (webcam or uploaded video). Teal: the rPPG signal chain. Amber: the learned correction layer. Coral: the deterministic clinical engine – the sole source of truth for a custom ESI-style acuity score, the Compensated Shock flag, and priority score. Pink: the CrewAI narration layer, which only restates and explains that output. Badges mark the two numbers motivating this separation: a 31% MAE reduction from the correction layer, and 6.2% narration faithfulness for the one field a prior revision let the LLM compute unsupervised (Sec. V-D).

respiratory rate exceeds 20 breaths/min, treating a barely-elevated reading the same as a severe crisis, severity above that gate is graded by a **multiplicative interaction**:

$$\text{tachy}_{HR} = \text{ramp}(HR,\ \text{threshold} = 100,\ \text{span} = 40) \quad (1)$$

$$\text{tachy}_{RR} = \text{ramp}(RR,\ \text{threshold} = 20,\ \text{span} = 15) \quad (2)$$

$$\text{shock}_{\text{interaction}} = \text{tachy}_{HR} \times \text{tachy}_{RR} \quad (3)$$

where $\text{ramp}(v, t, s)$ rises linearly from 0 to 1 as v climbs from t to $t + s$. Multiplying rather than averaging or OR-ing means risk accumulates only when both vitals are elevated together, matching the physiological definition of *compensated* shock as a combined cardiorespiratory response. HRV (RMSSD) and a **Baevsky Stress Index** are folded in as secondary, lower-weight terms, and any Compensated Shock flag unconditionally forces manual review. We describe this deliberately as a **custom, vitals-only scoring rule**, not a faithful ESI [7] or NEWS2 [8] implementation – both need inputs (chief complaint, SpO_2, blood pressure, temperature) a webcam cannot measure.

E. Multi-Agent Narration Layer

Three CrewAI agents run in sequence: a **Perception Agent** restates already-extracted vitals, a **Diagnostic Agent** explains the ESI level and shock flag using retrieved guideline context, and a **Coordinator Agent** restates the priority score. Guideline grounding uses **term-frequency cosine similarity**, simpler than comparable RAG pipelines [13]. Each request spins up a fresh crew, since CrewAI's task objects store output internally and are not safe to re-enter concurrently.

Why the LLM does not compute the ESI level. An earlier revision let the agents freely generate the ESI level and shock flag as JSON, falling back to a hardcoded default of 3 on any parse failure – making the most safety-critical number non-deterministic. Section V-D presents the measurement that motivated reverting this.

F. Dashboard Interface

Fig. 2 shows VITAL's two primary dashboard views: **Vitals Monitor** (top), exposing the live intake feed, optical physiology indicators, and rPPG waveform panel; and **Triage Dispatch** (bottom), the central queue sorted by acuity. Both are shown mid-session on genuine acquisitions – Vitals Monitor with a live face-ROI overlay and populated heart-rate, respiration, and stress-index readouts, and Triage Dispatch holding real acuity-sorted queue entries, including a Compensated Shock case at ESI level 1.

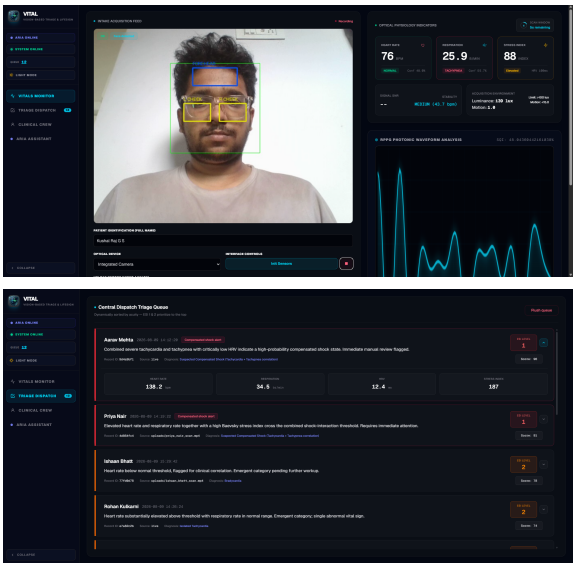


Fig. 2. VITAL's dashboard: Vitals Monitor (top) mid-acquisition, Triage Dispatch (bottom) populated with real session data.

IV. Dataset Description

Real-video validation draws on **71 sessions across 51 unique subjects**, pooled from two public datasets [4], [5]. The three DSP bugs (Section V) were root-caused on an initial **56-session** set; **15** more UBFC-Phys sessions, found later already on disk, were folded in through the same fixed pipeline. They were harder on average – the classical baseline rose from 8.42 to **9.43 BPM MAE** once included – so every Section VI number

reflects the harder pool, not the easier subset the bugs were found on.

- **UBFC-rPPG** [4] – **41 sessions** (both public releases combined), a math-stress protocol with ground truth from a contact pulse-oximeter waveform.
- **UBFC-Phys** [5] – **30 sessions** (10 subjects × 3 conditions: rest, speech, arithmetic, a Trier Social Stress Test-inspired protocol), ground truth from a wrist-worn BVP waveform, peak-detected into a per-session BPM.

Class composition is uneven by design (10 rest, 10 speech, 10 arithmetic, 41 unlabeled UBFC-rPPG), reported stratified rather than pooled into a misleading average (Table IV). Every model uses **GroupKFold by subject identifier** – no subject's sessions appear in both folds – keeping the correction model from **memorizing a specific person**.

V. Methodology

A. *Classical DSP Bug-Fixing, By Tracing Individual Failures*

Rather than tuning global thresholds against an aggregate metric, each bug below was found by tracing one specific failing subject to its root cause, then verified against the full real-video set. Table I summarizes the three bugs identified.

TABLE I
Classical DSP Bugs, Root-Caused and Fixed

Bug	Root Cause	Fix, Verified Effect (BPM)
Harmonic lock-on	Peak-picker locks onto half the true rate for sustained stretches	Promote to the double when it has comparable power; subj.41 52.6→8.2, subj.46 41.2→1.3, subj.9 60.9→2.8
Hardcoded frame rate	Every engine assumed 30 fps regardless of the real source rate	Read the true source fps; subj.25 23.5→3.5, subj.26 25.5→2.5
History freeze	An 18 BPM noise-rejection gate can anchor on an early wrong burst forever	Allow re-anchoring only while the anchor is still small; subj.37 21.0→6.0, no regression elsewhere

The history-freeze fix needed a second look: a first attempt fixed its target subject but **silently regressed the harmonic-lock-on fix's flagship case** to a 41.8 BPM error, revealing the earlier "fixed" result had been propped up by this same freeze bug anchoring correctly by luck, not design. A narrower fix resolved both without re-breaking either. Together, all three fixes moved the 56-session baseline from **17.4 to 8.4 BPM MAE (52% relative reduction)**; RMSE 24.1→15.6.

B. *Hybrid Classical-DSP + ML Correction*

A **gradient-boosted regressor** (`GradientBoostingRegressor`, **Huber loss**) corrects CHROM's per-cycle BPM output using **ten features per cycle**: raw BPM, signal-quality index, SNR, a stability score, respiratory rate and its confidence, HRV, the Baevsky stress index, and the adaptive filter's cutoff frequencies – each a causal, real-time output of the DSP engine, computed only up to that cycle, so none carries information from later in the session. The model is evaluated with **5-fold GroupKFold by subject**, so every session-level result in Section VI comes from a model that never saw that subject during training. Huber loss was adopted after comparing it against squared-error loss on the same folds, producing a small but consistent gain (**8.80→8.67 BPM** per-cycle MAE, **6.65→6.48 BPM** session-level MAE) – consistent with squared-error over-weighting a handful of extreme-outlier sessions. Larger models – a bigger regularized neural network, a bigger gradient-boosted model, an ensemble of both – **did not beat the small deployed model**; the binding constraint is **51 unique subjects**, not labeled-cycle count. One honest caveat: hyperparameters are fixed constants, never tuned inside the GroupKFold loop, but the **loss-function choice** was a form of **model-selection bias** – chosen by comparing performance on the same 71 sessions later reported as the final result.

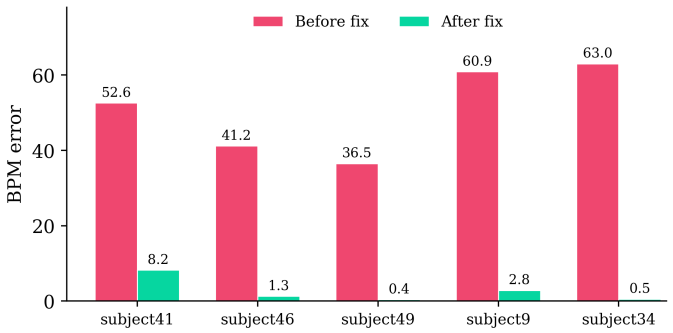


Fig. 3. Per-subject BPM error before and after the subharmonic-promotion fix for the harmonic lock-on bug. Zero effect on the shipped CHROM synthetic MAE (0.29, unchanged).

C. *POS's Own Bug, Partially Fixed*

POS [3] shows a **mirror-image failure to CHROM's**: it overshoots to roughly double the true rate from cold start. A symmetric "demote to the half" check cut POS's synthetic MAE from **26.71 to 16.00 BPM (28.88 RMSE)** with zero effect on CHROM – real but partial, two of seven synthetic cases still fail. On real video, tested here for the first time, results were mixed: one subject accurate (1.7 BPM error), two others failed oppositely, and no threshold fixed either – one subject's true rate never appears as a candidate, the other's is present but weak (median **11% power**, never crossing 30%). The cause is structural: CHROM applies global mean-normalization and variance equalization before projecting; POS applies only per-window mean-normalization [1]. Fixing this needs a rewrite of POS's projection math, so it stays **experimental**.

D. *Agent Narration Faithfulness Evaluation*

To test whether the LLM narration layer can be trusted with clinical numbers directly, we ran the real CrewAI pipeline – live API calls to an **8-billion-parameter** instruction-tuned LLM – against **8 controlled clinical scenarios** spanning the acuity range (normal, isolated tachycardia, isolated tachypnea, mild shock, severe shock, bradycardia, low signal-quality, no-signal), each run twice for **16 total runs**, checking whether the LLM's JSON output exactly matched the deterministic

assessment on every field despite instructions to restate values verbatim.

VI. Experimental Results

A. Synthetic Self-Test

Table II isolates the DSP math from camera capture and face detection, using a physiologically-modeled synthetic RGB trace across seven known-BPM trials (50-140 BPM, 2% sensor noise).

TABLE II
SYNTHETIC VALIDATION (DSP CORE ONLY)

Algorithm	MAE (BPM)	RMSE (BPM)
CHROM (shipped default)	**0.29**	0.53
POS (experimental, not wired in)	16.00	28.88

B. Real-Video Classical vs. Hybrid Accuracy

Table III reports the headline result across all 71 real sessions; the correction model is always tested on subjects it has never seen, so it cannot have simply memorized their patterns.

TABLE III
REAL-VIDEO ACCURACY, 71 SESSIONS / 51 SUBJECTS

Pipeline	MAE (BPM)	RMSE (BPM)
Classical DSP only (CHROM)	9.43	16.51
Hybrid (DSP + ML correction)	**6.48**	**9.66**
Relative MAE reduction	31%	

TABLE IV
PER-CONDITION ACCURACY BREAKDOWN

Condition	Classical	Hybrid	n
Speech (talking)	21.0	**7.8**	10
Arithmetic (mental stress)	10.4	**7.0**	10
UBFC-rPPG (unlabeled task)	7.4	6.1	41
Rest	5.1	6.1 (worse)	10

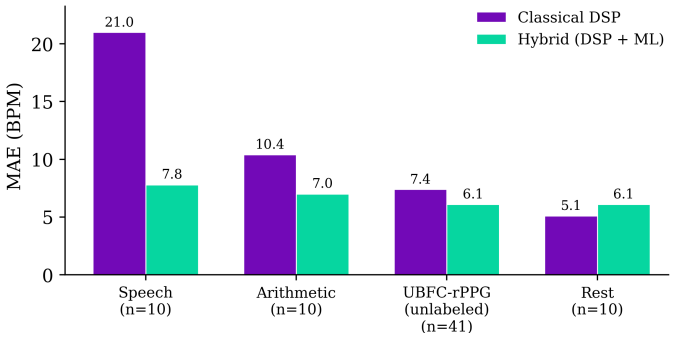


Fig. 4. Classical vs. hybrid MAE by condition. The hybrid model's gain concentrates in speech and arithmetic; on already-clean rest it is slightly worse, since little systematic bias remains for it to correct.

The hybrid model helps every condition except already-clean rest, where the classical baseline has little bias left to correct (Fig. 4). **Its largest single gain is in speech** – the one condition three independent DSP-level fixes never touched, since mouth-movement motion artifact is a fundamentally different failure mode from frequency-domain harmonic ambiguity or a frame-rate mismatch.

C. Subject-Level Significance

The **31% relative MAE reduction** is real, but not uniform. Averaging each subject's sessions first, only **30 of 51 subjects** show any net improvement; the bootstrap **95% CI** on the mean subject-level gain is **[-0.19, 4.54] BPM**, crossing zero, and a paired Wilcoxon test does not reach significance (**p=0.24**). The top 5 subjects alone, corrected by 15-36 BPM each, account for **101.7%** of the total gain – the rest roughly cancel out. The correction layer's value is rescuing specific hard failures, not improving every session; we report this rather than let the aggregate imply otherwise.

D. Comparison Against the Closest Published Benchmark

Table V restricts the comparison to the UBFC-rPPG-only subset directly comparable to Qaadan et al.'s evaluation [1], since their study does not include UBFC-Phys. Their evaluation uses a single 80/20 split, not subject-level GroupKFold, so these values are best read as reported benchmark context, not a strictly controlled head-to-head comparison.

TABLE V
COMPARISON ON THE UBFC-RPPG-ONLY SUBSET

Method	MAE (BPM)
This work – classical DSP only	7.42
This work – hybrid (DSP + ML correction)	6.12
Qaadan et al. – CHROM + XGBoost (their best) [1]	**4.60**

We report this plainly: on rPPG accuracy alone, **their approach beats both of ours**, not by a rounding margin. Their permutation-importance analysis ranks **skewness, lag-1 autocorrelation, sample entropy, and spectral centroid** among their top features, none of which appear in VITAL's ten-feature schema – a concrete, unclosed gap. What VITAL has, that a pure benchmarking study does not, is the rest of this section.

E. Agent Narration Faithfulness

Table VI reports the field-level exact-match rate between the LLM's restated output and the deterministic ground truth it was given, across all 16 runs.

TABLE VI
LLM AGENT NARRATION FAITHFULNESS (16 RUNS, 8B-PARAMETER LLM)

Field	Faithfulness rate
JSON parses at all	15/16 (93.8%)
ESI level restated correctly	13/16 (81.2%)
Priority score restated correctly	**1/16 (6.2%)**
Shock flag restated correctly	15/16 (93.8%)
Diagnosis text – exact wording	6/16 (37.5%)
Diagnosis text – correct meaning, different words	12/16 (75.0%)

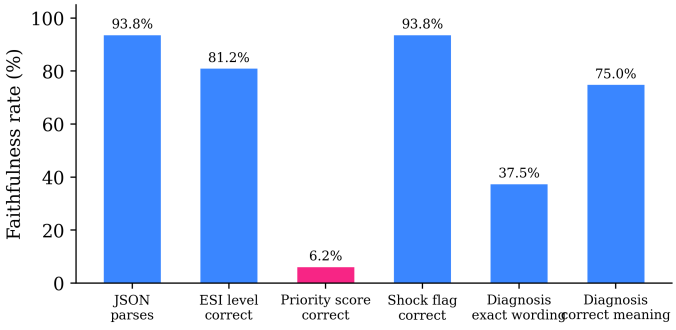


Fig. 5. LLM agent narration faithfulness by field, 16 runs. The priority score (highlighted) is the one field an earlier system revision let the LLM compute unsupervised; it is also the field the LLM restates least reliably.

Even told to restate rather than recompute, the priority-score field was rarely a plausible miss. In **9 of 16 runs** the LLM substituted the patient's own **ESI level integer** into that field (e.g. `2` for a true score of **58**) – a repeatable **field-confusion error**, not drift; the rest mixed stray decimals with non-numeric text (`"Resuscitation"`, `"Emergent"`). Checking whether the true value appeared anywhere in the raw output, to rule out a schema artifact, still gave **1/16 (6.2%)** – a substituted wrong value, not a formatting slip. The LLM also called tools that do not exist, a second gap. This is a small, 16-run sample under one 8B configuration: a failure of **that configuration**, not LLMs generally.

VII. Discussion

A. What the Numbers Actually Mean

Table VI's result shows this safety rule was **necessary, not merely cautious**. Trusting the LLM's own JSON for the priority score would have produced an incorrect priority-score value in **more than 9 of every 10 runs**. Comparable agentic-orchestration architectures guard against this failure procedurally, without measuring how often an agent's output diverges from ground truth [14]; this is what that divergence looks like, measured.

The hybrid model's largest gain sits where DSP fixes could not reach: speech's motion artifact is **orthogonal** to harmonic ambiguity and frame-rate mismatch, so a learned correction absorbs a bias no heuristic can express – also why bigger models did not help, since the real limit is **51 subjects**, not cycle volume.

B. Positioning and Limitations

Not an accuracy paper competing on a single metric – Qaadan et al.'s is a pure rPPG-accuracy benchmark with no deployed system, clinical-safety layer, or agent-reliability measurement, and Table V shows they win on their own metric. Real limits remain: **no real-data respiratory-rate accuracy** (unvalidated RR feeds both the ML features and shock-interaction term); **no skin-tone-stratified accuracy**; a **single-session** live-camera path; and a **custom, vitals-only** rule, not validated ESI or NEWS2 [7], [8]. To be precise: this paper validates **rPPG accuracy** and **LLM faithfulness**, not clinical triage accuracy.

VIII. Conclusion and Future Work

VITAL demonstrates a contactless vitals-to-triage pipeline: a hybrid correction layer **closes 31% of the real-video accuracy gap** left by classical rPPG (**9.43→6.48 BPM MAE**, 71 sessions) – real, concentrated in a subset of hard subjects, not a broad effect – and a deterministic clinical engine stays the **sole source of truth** for triage-critical numbers, justified by measuring that the LLM narration layer restates priority score correctly only **6.2% of the time**. Three root-caused DSP bugs, one partial POS fix, and the failed fixes are reported alongside a real gap against the closest benchmark, not hidden.

Future work: closing the **named feature gap** against [1] (skewness, autocorrelation, sample entropy, spectral centroid); a **structural rewrite** of POS's projection step; **skin-tone-stratified validation** once compatible metadata exists; and a **live, real-patient pilot**, out of scope here but planned. Also open: a **nested-GroupKFold** Huber re-evaluation, and **expanding the LLM faithfulness run** beyond 16 trials.

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